

M.H. OSHADHA CANCHANA

INTERN BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

Third-year BSc (Hons) Business Information Systems undergraduate at the University of Sri Jayewardenepura (Current GPA: 3.87/4.00), seeking a Backend Developer Internship. Experienced in engineering robust RESTful APIs, transactional database architectures, and microservices using Python, FastAPI, Django REST Framework, Flask, and PostgreSQL. Demonstrated expertise in relational database design, query optimization, Role-Based Access Control (RBAC), and integrating intelligent AI/LLM pipelines (MCP, RAG, LangGraph, Vision models) into scalable backend workflows.

TECHNICAL SKILLS

Programming Languages: Python, Java, C#, SQL, JavaScript

Backend & API Frameworks: FastAPI, Django REST Framework (DRF), Flask, Spring Boot, RESTful APIs, Microservices Architecture

Databases: PostgreSQL, MySQL, SQLite, MongoDB, ChromaDB

AI & Agentic Systems: Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Google Gemini Vision

Machine Learning: Scikit-learn, XGBoost, Random Forest

DevOps & Developer Tools: Docker, CI/CD Pipelines, Git, GitHub, Postman

TECHNICAL PROJECTS

End-to-End Election Management System | *Django REST Framework, PostgreSQL, REST APIs* **Jun 2026 – Jul 2026**

Links: [Live Demo](#) | [GitHub Repository](#)

- **Architected a secure, transactional backend** for election administration using Django REST Framework and PostgreSQL, enforcing strict data isolation and zero vote leakage.
- **Implemented Role-Based Access Control (RBAC)** and atomic database transactions to maintain vote integrity, prevent concurrency conflicts, and eliminate duplicate voting attempts.
- **Designed normalized 3NF database schemas** and optimized REST API endpoints for candidate profiles, ballot generation, voter authentication, and real-time election tallying.

Hybrid-Retrieval Research Assistant | *Fast API, Lang Graph, Chroma DB, SQLite FTS5, Doling* **Aug 2026-Present**

Links: [GitHub Repository](#)

- **Engineered a high-performance FastAPI backend** for an autonomous multi-agent research assistant that streamlines document parsing, retrieval, and synthesis.
- **Implemented hybrid search architecture** combining ChromaDB dense vector embeddings with SQLite FTS5 BM25 sparse keyword retrieval, unified via Reciprocal Rank Fusion (RRF).
- **Built automated document processing pipelines** utilizing Docling and LangGraph state machines for reliable document chunking, metadata extraction, and multi-step reasoning.

Automated MCQ & Visual Element Extraction API | *Flask, Google Gemini Vision, REST APIs* **Feb 2026 – Mar 2026**

Links: [GitHub Repository](#)

- **Developed an intelligent REST API in Flask** to parse complex exam papers, automatically extracting questions, options, formulas, and diagrams from raw PDFs.
- **Integrated Google Gemini Vision API** into backend ingestion pipelines for high-accuracy bounding box detection, visual element classification, and OCR extraction.
- **Designed strict structured JSON output schemas** to ensure seamless data interoperability and zero-error ingestion into downstream Learning Management Systems.

Machine Learning Prediction & Inference Service | *Flask, XGBoost, Random Forest, Scikit-learn, Pandas* **Sep 2025 – Nov 2025**

Links: [GitHub Repository](#)

- **Constructed a production-ready Flask REST service** to serve real-time predictions from trained ensemble regression models (XGBoost and Random Forest).
- **Implemented end-to-end data pipelines** for automated feature scaling, data cleaning, and model evaluation, optimizing Mean Absolute Error (MAE).

WORK EXPERIENCE

Audit Intern | *Jayasinghe & Company* | *94/12, 2nd Lane, Kirulapone Ave*

May 2023 – Mar 2024

Colombo, Sri Lanka

- **Prepared comprehensive financial audit schedules**, working papers, and statutory compliance documentation across diverse client enterprise portfolios.
- **Conducted rigorous ledger reconciliations** and substantive analytical procedures to identify data anomalies and ensure 100% financial reporting integrity.
- **Evaluated internal accounting controls** and executed transactional testing to verify data accuracy, completeness, and regulatory compliance.

EDUCATION

BSc (Hons) in Business Information Systems

Feb 2024 – Present

University of Sri Jayewardenepura | *Third Year Standing*

Current GPA: 3.87 / 4.00

- **Relevant Coursework:** Database Management Systems, Data Structures & Algorithms, Object-Oriented Programming (OOP), Enterprise Systems Development, Web Application Development.

G.C.E. Advanced Level – Commerce Stream

2009 – 2023

Royal College, Colombo 07

- **Academic Results:** 3 Distinctions (A) in Business Studies, Accounting, and Economics | Z-Score: 2.087
- **National Rankings:** Island Rank: 304 | Colombo District Rank: 96

PROFESSIONAL QUALIFICATIONS

Chartered Accountancy – CA Sri Lanka

2023 – Present

The Institute of Chartered Accountants of Sri Lanka | *Corporate Level (Active Student)*

HONORS & AWARDS

4th Place Worldwide & Lex Fantastic Stage Winner

Apr 2026

MaxUp 2026 International Legathon – Global Finals | *Astana, Kazakhstan*

- **International Achievement:** Represented Sri Lanka at the Global Finals, ranking 5th in preliminary rounds and finishing 4th overall worldwide while winning the Lex Fantastic Stage debate championship against Georgia.
- CA Sri Lanka National Subject Prize Winner** **2024 – 2025**
- **Business Law II (BL 5) National Subject Prize:** Awarded for achieving the highest island-wide score in the September 2025 Examination.
 - **Business Law I (BL 6) National Subject Prize:** Awarded for achieving the highest island-wide score in the March 2024 Examination.